

Machine Learning – Chapter 2: Understanding Data

Based on S. Sridhar & M. Vijayalakshmi, Machine Learning, Oxford University Press

Detailed teaching slides • Data understanding, preprocessing, statistics, probability and feature engineering

Chapter 2 – Understanding Data

Data is the foundation on which machine-learning algorithms operate.

Before selecting or applying an ML algorithm, the dataset must be understood, represented, cleaned, summarized and explored.

This chapter covers data types, data management, analytics, visualization, statistics, multivariate analysis, probability, hypothesis testing, and feature engineering.

Learning Objectives

Understand data and different types of data.

Understand data storage and representation.

Learn the basics of data management and preprocessing.

Understand descriptive statistics and visualization.

Perform univariate, bivariate and multivariate data analysis.

Review essential mathematics, probability and statistical testing.

Understand feature engineering and dimensionality reduction.

2.1 What is Data?

Data consists of facts that can be processed by a computer.

Examples include numbers, text, images, audio, video and other digital representations.

Data may be stored in files, databases or data warehouses.

Operational data is generated during normal business processes; non-operational data is often used for information and decision-making.

Data becomes useful for ML only after appropriate preparation and interpretation.

Elements / Characteristics of Big Data – The 6 Vs

Volume – amount of data generated and stored.

Velocity – speed at which data is generated and processed.

Variety – diversity of forms and sources: text, image, video, audio, etc.

Veracity – reliability, truthfulness and quality of data.

Validity – correctness or suitability of data for the intended task.

Value – useful information extracted from data and its influence on decisions.

Data Quality

Data quality strongly affects the quality of analysis and ML models.

Numeric data quality can be considered through factors such as precision and accuracy.

Precision concerns closeness among repeated measurements.

Accuracy concerns closeness of a measurement to the true value.

Poor quality data can produce misleading patterns and unreliable models.

2.1.1 Types of Data

Structured data – organized in a predefined structure such as tables.

Unstructured data – data without a fixed tabular structure, including images, audio, text documents and programs.

Semi-structured data – partly organized data such as XML/JSON, RSS feeds, hierarchical data.

Structured data can further be represented as records, data matrices, graphs and ordered data.

Structured Data

Record data: measurements organized as objects/records with attributes or fields.

Data matrix: numerical attributes arranged as rows and columns.

Graph data: represents relationships among objects.

Ordered data: attributes have an implicit order.

Examples of ordered data include temporal, sequence and spatial data.

Ordered Data

Temporal data – observations associated with time; e.g., temperature measured over time.

Sequence data – ordered symbols/items without necessarily having timestamps.

Spatial data – observations associated with positions, regions or locations.

The ordering itself contains useful information and should be preserved during analysis.

2.1.2 Data Storage and Representation

After data collection, data must be stored in a form suitable for analysis.

Flat files are simple and commonly used for small datasets.

Common tabular formats include CSV and TSV.

Database systems support storage, querying, transactions and administration.

The chapter also discusses spatial databases, WWW, XML, data streams, RSS and JSON.

Database and Other Representations

Transactional databases store records associated with transactions.

Time-series databases store time-indexed measurements and events.

Spatial databases store geographic/spatial information.

The World Wide Web is a major online information source.

XML and JSON provide structured/semi-structured data interchange; data streams represent continuously arriving data.

2.2 Big Data Analytics and Types of Analytics

Data analytics assists organizations in making decisions.

Data analysis and data analytics are related but not identical: analytics emphasizes extracting useful insights and decision support.

The chapter identifies four types:

1. Descriptive analytics – describing the main features of the data
 2. Diagnostic analytics- causal analysis , (deals with 'why') finds the cause and effect of the events
 3. Predictive analytics- deals with the future. "What will happen in future given the data"
 4. Prescriptive analytics- finding the best course of action for the business organizations.
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Four Types of Analytics

Descriptive – What happened? Summarizes and describes collected data. Data aggregation, data summarization, statistical measures, frequency distributions, data visualization, dashboards, reports, mean/median/mode, standard deviation

Diagnostic – Why did it happen? Looks for causes/effects and associated factors. Drill-down analysis, correlation analysis, root-cause analysis, variance analysis, comparative analysis, association analysis, hypothesis testing

Predictive – What will happen? Uses patterns in historical data to predict future outcomes. Regression, classification, time-series forecasting, decision trees, random forests, neural networks, clustering, machine learning models

Prescriptive – What should be done? Goes beyond prediction to recommend actions. Optimization, simulation, what-if analysis, decision analysis, rule-based systems, constraint optimization, reinforcement learning, recommendation systems

2.3 Big Data Analysis Framework

The chapter presents a four-layer analytics architecture:

1. Data connection layer – data ingestion and connectors; includes ETL activities.
2. Data management layer – storage, retrieval, querying and parallel data management.
3. Data analytic layer – statistical tests, ML algorithms, model construction and validation.
4. Presentation layer – dashboards and applications for displaying results.

Data Collection

Data collection is the first step in the big-data processing cycle.

Good-quality data should satisfy three important properties:

- Timeliness – relevant and not obsolete.

- Relevancy – directly useful for the ML/data-mining problem and sufficiently representative.

- Knowledge about the data – understandable and interpretable by domain experts.

Sources discussed include open/public data, social media and multimedia data.

2.3.2 Data Preprocessing

Real-world data is often 'dirty'.

Common problems include incomplete data, inaccurate data, outliers, missing values, inconsistent values and duplicate data.

Data cleaning identifies and corrects/removes errors before modeling.

Cleaning may include handling missing values, smoothing noise, removing outliers and resolving inconsistencies.

Missing Data Analysis

Common strategies described in the chapter:

1. Ignore the tuple when the missing value is associated with the class label.
2. Fill manually when feasible and domain knowledge is available.
3. Use a global constant such as 'Unknown' or a specified value.
4. Use the attribute average.
5. Use the class-specific average.
6. Use the most probable value inferred from other methods/classification.

Outliers and Noise

Noise is random error or variance in measured data.

Outliers are observations that differ markedly from the rest of the data.

Binning can be used to smooth noisy data.

The chapter illustrates smoothing by bin means and smoothing by bin boundaries.

Outlier handling should be driven by domain knowledge and the purpose of the analysis.

Binning Example

Dataset: 12, 14, 19, 22, 24, 26, 28, 31, 34.

With equal-frequency binning of size 3:

Bin 1 = 12, 14, 19

Bin 2 = 22, 24, 26

Bin 3 = 28, 31, 34

Bin means can replace values within each bin; boundaries can replace values by the nearest bin boundary.

Data Integration and Transformation

Data integration combines data from multiple sources into a single data source.

The aim is to merge data and remove redundancies introduced during integration.

Data transformation changes data into a form appropriate for ML algorithms.

Normalization scales attributes to comparable ranges.

The chapter introduces Min–Max normalization and z-score normalization.

Min–Max Normalization

Maps a value V from its original range $[\min, \max]$ to a target range $[\text{new_min}, \text{new_max}]$.

Formula:

$$V' = ((V - \min) / (\max - \min)) \times (\text{new_max} - \text{new_min}) + \text{new_min}$$

For a 0–1 target range, the transformation becomes $(V - \min)/(\max - \min)$.

Useful when features have different numeric scales.

z-Score Normalization

Uses the mean and standard deviation of an attribute.

Formula: $z = (V - \mu) / \sigma$

The transformed variable is centered around zero and measured in standard-deviation units.

The chapter notes its use for detecting possible outliers.

Example in the text: values 10, 20, 30 produce z-scores -1, 0 and +1 for $\mu = 20$ and $\sigma = 10$.

2.4 Descriptive Statistics

Descriptive statistics summarizes and describes a dataset.

It is descriptive: it does not by itself go beyond the observed data.

It supports understanding of data before applying ML algorithms.

Exploratory Data Analysis (EDA) uses descriptive statistics and visualization to investigate data.

Key areas: data types, central tendency, dispersion and shape.

Measurement / Data Types

Categorical (qualitative) data:

- Nominal – categories without meaningful order.

- Ordinal – categories with an order.

Numerical (quantitative) data:

- Interval – meaningful differences; zero is not an absolute absence.

- Ratio – meaningful differences and ratios; zero has a meaningful interpretation.

Data may also be discrete or continuous.

Discrete, Continuous, Univariate, Bivariate, Multivariate

Discrete data: countable values, typically recorded as integers.

Continuous data: can take values over a range, including decimal values.

Univariate data: one variable.

Bivariate data: two variables.

Multivariate data: more than two variables.

2.5 Univariate Data Analysis and Visualization

Univariate analysis studies one variable at a time.

It focuses on frequency distribution, central tendency, dispersion and shape.

Visualization helps present, summarize and explore data.

Chapter examples include bar charts, pie charts, histograms and dot plots.

Visualization – Bar, Pie, Histogram, Dot Plot

Bar chart – displays frequency/counts for categories or discrete values.

Pie chart – shows parts of a whole using percentage/proportion.

Histogram – groups numerical values into intervals/bins and shows frequency distribution.

Dot plot – displays individual observations as points and is less clustered than a bar chart.

Choice of visualization should match the variable type and analysis objective.

Central Tendency

Mean – arithmetic average: $\bar{x} = (1/N) \sum x_i$.

Weighted mean – observations contribute according to their weights.

Geometric mean – Nth root of the product of N values.

Median – middle value after sorting; useful when data contains extreme values.

Mode – most frequently occurring value; particularly useful for discrete/categorical data.

Dispersion

Range = maximum – minimum.

Standard deviation measures spread around the mean.

Quartiles divide ordered data; Q1, Q2 (median) and Q3 are important.

Interquartile range: $IQR = Q3 - Q1$.

The chapter identifies values above $Q3 + 1.5 \times IQR$ or below $Q1 - 1.5 \times IQR$ as potential outliers.

Semi-interquartile range = $1/2 IQR$.

Five-Point Summary and Box Plot

Five-point summary: minimum, Q1, median (Q2), Q3 and maximum.

Box plot visually represents these five statistics.

The box represents the middle 50% of observations.

The median is shown inside the box.

Whiskers extend toward the lower and upper extremes; the plot helps reveal spread and possible outliers.

Shape of Data – Skewness and Kurtosis

Skewness describes asymmetry of a distribution.

Positive skew: longer tail toward higher values; mean is generally greater than median.

Negative skew: longer tail toward lower values; mean is generally less than median.

Kurtosis describes the peakedness/tail behavior relative to a normal distribution.

The chapter also discusses mean absolute deviation (MAD) and coefficient of variation (CV).

Special Univariate Plots

Stem-and-leaf plot preserves individual observations while showing distribution shape.

A Q–Q plot compares sample quantiles with theoretical normal quantiles.

If points approximately follow the reference line, the data may be reasonably consistent with the theoretical distribution.

Large deviations from the line suggest a different distributional shape.

2.6 Bivariate Data

Bivariate data contains two variables.

The purpose is to identify relationships between variables and explore possible causes/effects.

The chapter uses temperature and sweater sales as an example.

Scatter plots are used to visualize the relationship.

Line graphs can also show trends in paired observations.

Covariance

Covariance measures the joint variability of two variables X and Y.

Population form: $\text{Cov}(X,Y) = (1/N) \sum [(x_i - E(X))(y_i - E(Y))]$.

Positive covariance: variables tend to increase together.

Negative covariance: one tends to increase when the other decreases.

The magnitude depends on the measurement scales, so covariance is not directly standardized.

Pearson Correlation

Correlation standardizes covariance and measures the direction and strength of linear association.

$$r = \text{Cov}(X,Y)/(\sigma_X \sigma_Y).$$

r is bounded between -1 and $+1$.

Positive $r \rightarrow$ positive linear relationship.

Negative $r \rightarrow$ negative linear relationship.

r near zero indicates little/no linear relationship; it does not by itself prove independence.

2.7 Multivariate Statistics

Multivariate data involves more than two variables.

The objective is to study relationships among several variables simultaneously.

The chapter mentions regression analysis, principal component analysis and multivariate analysis of variance as examples.

A multivariate dataset can be represented as a matrix of observations and attributes.

Mean becomes a mean vector; covariance becomes a covariance matrix.

Heatmap and Pairplot

Heatmap is a graphical representation of a 2D matrix using color intensity.

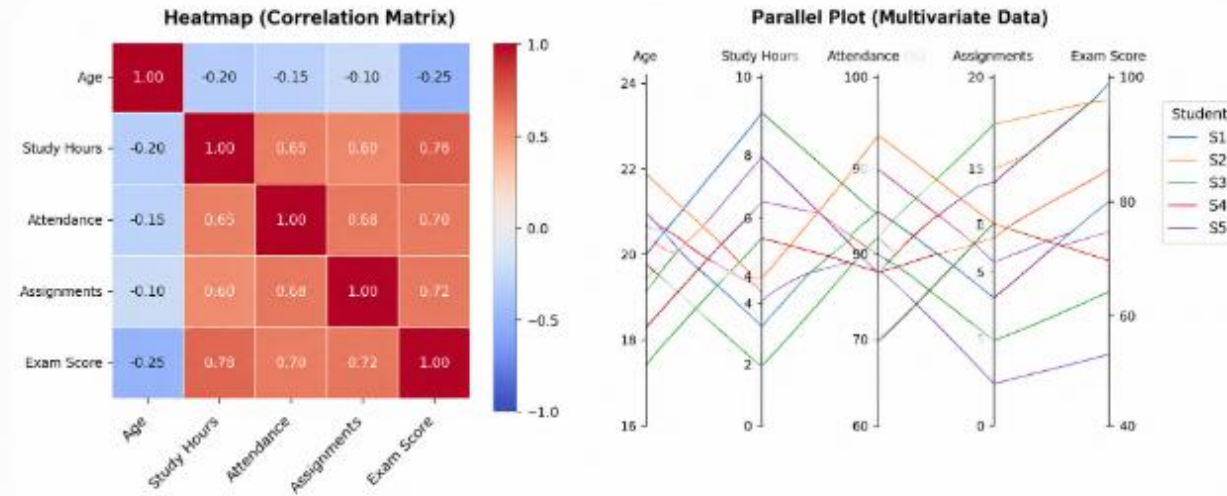
It can make patterns in large matrices easier to perceive.

Pairplot/scatter matrix displays pairwise scatter plots for multiple variables.

Pairplots help visually inspect relationships, trends, clusters and possible correlations.

These are exploratory tools; interpretation should be supported by numerical measures.

Correlation Heatmap and Student Parallel Plot



2.8 Essential Mathematics for Multivariate Data

Machine learning uses concepts from linear algebra, statistics and probability.

Linear algebra provides tools for vectors, matrices and transformations.

The chapter introduces linear systems, Gaussian elimination and matrix decomposition.

Probability distributions support probabilistic modeling and statistical reasoning.

Linear Systems and Gaussian Elimination

A linear system can be represented as $Ax = y$.

For a nonsingular square matrix, x can be written as $A^{-1}y$.

Gaussian elimination transforms the augmented matrix into reduced/echelon form.

Basic row operations: swap rows; multiply/divide a row by a constant; add/subtract a multiple of another row.

Back-substitution is used after elimination to obtain remaining unknowns.

Matrix Decompositions – LU

Matrix decomposition expresses a matrix as a product of simpler matrices.

LU decomposition represents A as $A = LU$.

L is a lower triangular matrix.

U is an upper triangular matrix.

Gaussian elimination can be used to obtain L and U .

Decomposition is useful when repeated or complex matrix operations are required.

Probability Distributions

A probability distribution describes how probability is associated with values of a random variable.

Two broad categories are discrete and continuous distributions.

PDF describes probability density for continuous variables.

CDF gives cumulative probability up to a value.

The chapter discusses Normal, Uniform, Exponential, Binomial, Poisson and Bernoulli distributions.

Continuous Distributions

Normal distribution – bell-shaped continuous distribution characterized by mean μ and standard deviation σ .

z-score: $z = (x - \mu)/\sigma$.

Uniform distribution – values in a specified interval have equal density.

Exponential distribution – models waiting/time-between-events in appropriate settings; parameter λ is used.

The correct distributional assumption matters in statistical modeling.

Discrete Distributions

Binomial distribution – models a fixed number of trials with two outcomes, such as success/failure.

Poisson distribution – models counts of events occurring over an interval under appropriate assumptions.

Bernoulli distribution – a single binary trial with outcomes represented by success/failure.

These distributions provide probability models for different types of ML/statistical problems.

Density Estimation and EM

Density estimation estimates the probability density function from observed data.

Parametric density estimation assumes a known distribution family and estimates its parameters.

Maximum Likelihood Estimation (MLE) chooses parameter values that maximize the likelihood of observed data.

The chapter introduces Gaussian Mixture Models and Expectation-Maximization (EM).

EM alternates between an expectation step and a maximization step until the parameters fit the latent-variable model.

Non-Parametric Density Estimation

Non-parametric methods do not assume a fixed parametric distribution.

Parzen window estimates density by counting observations within a region/window.

K-Nearest Neighbour (KNN) estimation uses the neighbourhood of a point to estimate density.

These methods can adapt to data distributions without specifying a particular parametric family.

2.9 Overview of Hypothesis Testing

A hypothesis is a statement about a population or data-generating process.

Null hypothesis H_0 is tested against an alternative hypothesis H_1 .

Statistical testing helps determine whether observed differences are statistically significant.

Typical steps: formulate hypotheses $\rightarrow$ select test/statistic $\rightarrow$ choose significance level $\alpha \rightarrow$ compute p-value $\rightarrow$ make a decision.

Type I error: reject a true H_0 . Type II error: fail to reject a false H_0 .

p-value and Confidence Interval

p-value indicates how compatible the observed result is with the null hypothesis under the test assumptions.

A small p-value provides evidence against H_0 .

Confidence interval gives a range of plausible values at a specified confidence level.

The chapter presents confidence level = $1 - \text{significance level}$.

Statistical conclusions should consider assumptions, sample size and practical significance.

Comparing Learning Methods – z and t Tests

z-test: used when assumptions such as known population variation and sufficiently large sample size are appropriate.

t-test: used to test means, especially when population variance is unknown and sample sizes are small.

One-sample t-test compares a sample mean with a theoretical/population mean.

Independent two-sample t-test compares means of two groups.

Paired t-test evaluates paired observations, such as before/after measurements.

Chi-Square Test

A non-parametric test used for categorical data.

The chapter uses a goodness-of-fit/association-style example involving observed and expected frequencies.

Statistic: $\chi^2 = \sum [(O_i - E_i)^2 / E_i]$.

Large discrepancies between observed and expected frequencies provide evidence against H_0 .

Degrees of freedom depend on the number of categories and constraints.

2.10 Feature Engineering

Features are attributes used by an ML model.

Feature engineering determines a useful subset or representation of features that can improve model performance.

Feature Transformation changes features into forms that may be more useful for learning.

Feature Selection identifies important features and removes irrelevant or redundant ones.

The chapter uses BMI as an example of creating a feature from height and weight.

Dimensionality Reduction

Feature selection reduces the number of attributes by retaining useful features.

With n features, there can be 2^n possible subsets, making exhaustive search expensive.

Reducing dimensionality can reduce computation and remove irrelevant information.

Feature selection is often a practical approach because it keeps the original feature representation.

The chapter introduces dimensionality reduction as part of feature engineering.

End-to-End Data Understanding Workflow

1. Collect relevant, timely and understandable data.
2. Identify data types and storage/representation.
3. Inspect data quality and detect missing values/outliers/inconsistencies.
4. Clean, integrate, transform and normalize data.
5. Explore with descriptive statistics and visualizations.
6. Study relationships using bivariate/multivariate analysis.
7. Apply appropriate probability/statistical methods.
8. Engineer/select features before ML modeling.

Quick Revision – Key Formulas

Mean: $\bar{x} = (1/N) \sum x_i$

Min–Max: $V' = ((V - \min) / (\max - \min))(\text{new_max} - \text{new_min}) + \text{new_min}$

z-score: $z = (V - \mu) / \sigma$

IQR = $Q3 - Q1$

$\text{Cov}(X, Y) = (1/N) \sum [(x_i - E(X))(y_i - E(Y))]$

Pearson $r = \text{Cov}(X, Y) / (\sigma_X \sigma_Y)$

Chi-square: $\chi^2 = \sum [(O_i - E_i)^2 / E_i]$

Exam-Oriented Questions

Define structured, unstructured and semi-structured data with examples.

Explain the 6 Vs of big data.

Explain the four types of analytics with examples.

Explain data preprocessing and methods for handling missing values/outliers.

Compare Min–Max and z-score normalization.

Explain mean, median, mode, standard deviation, IQR and box plot.

Differentiate covariance and correlation.

Explain Gaussian elimination and LU decomposition.

Explain normal, binomial, Poisson and Bernoulli distributions.

Explain hypothesis testing, p-value, z-test, t-test and chi-square test.

Explain feature engineering, feature selection and dimensionality reduction.

Chapter 2 – Takeaways

Good ML starts with good understanding of data.

Data type determines appropriate representation, preprocessing and visualization.

Cleaning and transformation improve the quality and suitability of data.

Descriptive, bivariate and multivariate analysis reveal structure and relationships.

Probability, statistics and linear algebra provide mathematical foundations for ML.

Feature engineering and dimensionality reduction prepare data for effective learning.